TURBOCOR-CLASS TWO-STAGE MAGNETIC-BEARING CENTRIFUGAL COMPRESSOR

Study model for SolidWorks + Ansys CFD | Basis: Danfoss Turbocor TTS350 as used in the Trane TCAA air-cooled chiller (IOM TCA-SVX001C-EN)

DESIGN POINT (from scripts/meanline.py)

Parameter	Value
Refrigerant	R-513A (R-1234yf/R-134a 56/44 wt%)
Capacity per compressor	110 tons (387 kW)
Sat. suction / discharge	3.5 C / 51.5 C
Suction / discharge pressure	364 / 1420 kPa(a)
Overall pressure ratio	3.90
Mass flow	3.34 kg/s
Suction volume flow	0.177 m3/s
Design speed	30,000 rpm
Shaft power / torque	113 kW / 35.9 N.m
Compressor kW/ton (est.)	1.08
Discharge temperature	60 C
Impeller tip dia. (both stages)	103.4 mm, tip speed 162 m/s
Blades	7 main + 7 splitter, backsweep 40 deg
Stage PR (1 / 2)	1.979 / 1.970

DRAWING INDEX

DWG	Title
TC-000	Cover sheet & design basis
TC-100/1	General arrangement - longitudinal section
TC-100/2	General arrangement - exterior views
TC-101	Impeller, stage 1
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TC-103	Rotor shaft assembly
TC-110	Inlet housing & inlet guide vanes
TC-120	Compressor casing flow path, de-swirl vanes
TC-121	Volute
TC-130	Motor stator & housing
TC-140	Magnetic & touchdown bearings
CFD-001	CFD fluid domains & boundary conditions

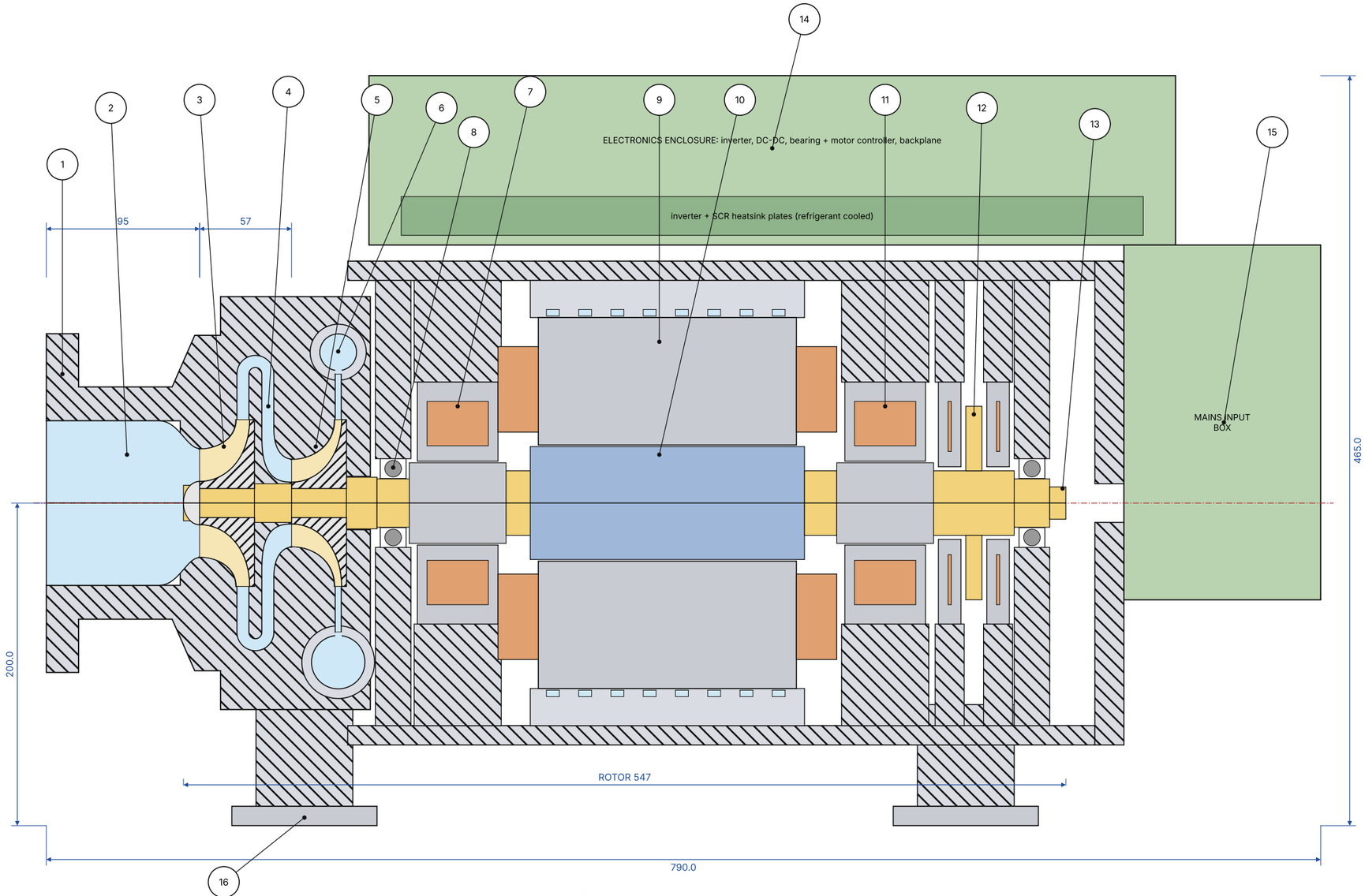
NOTES

1. All dimensions mm. Coordinate system: X = rotor axis, X=0 at suction flange face, +X toward motor.
2. Impeller/flow-path coordinates are generated - import them from cad_curves/ rather than re-typing.
3. Geometry is a physics-based preliminary design (meanline), proportioned on Turbocor TT-class
4. machines. It is NOT the OEM design and must not be used for manufacture or service parts.
5. Re-run scripts/ to change design point (speed, tonnage, refrigerant) - every sheet regenerates.
6. Materials: impellers 7075-T6 / 2618-T61 Al; casing A356-T6 Al; shaft 17-4PH; rotor sleeve CFRP;
7. laminations M270-35A / Co-Fe for bearings; housing A356-T6 / ductile iron.

Reverse-engineered study model - NOT OEM geometry. Third-angle projection. General tolerance +/-0.5 mm unless noted.

COVER SHEET & DESIGN BASIS

Turbocor-class 2-stage maglev centrifugal compressor (R-513A)	SHEET 1 OF 12
DWG TC-000 SCALE - UNITS mm	REV A 2026-10-17



ITEM LIST

#	Component
1	Suction flange DN100
2	Inlet guide vanes (8)
3	Impeller stage 1
4	De-swirl / return channel
5	Impeller stage 2
6	Volute
7	Front radial mag. bearing
8	Touchdown bearing (front)
9	Motor stator (cooling grooves)
10	PM rotor, CFRP sleeve
11	Rear radial mag. bearing
12	Axial (thrust) bearing
13	Axial sensor / rear touchdown
14	Electronics enclosure
15	Mains input box
16	Mounting feet

Section through rotor axis. Volute shown at 180 deg (top) and 360 deg (bottom). Gas: suction → IGV → stage 1 → diffuser → crossover → de-swirl vanes → stage 2 → volute.

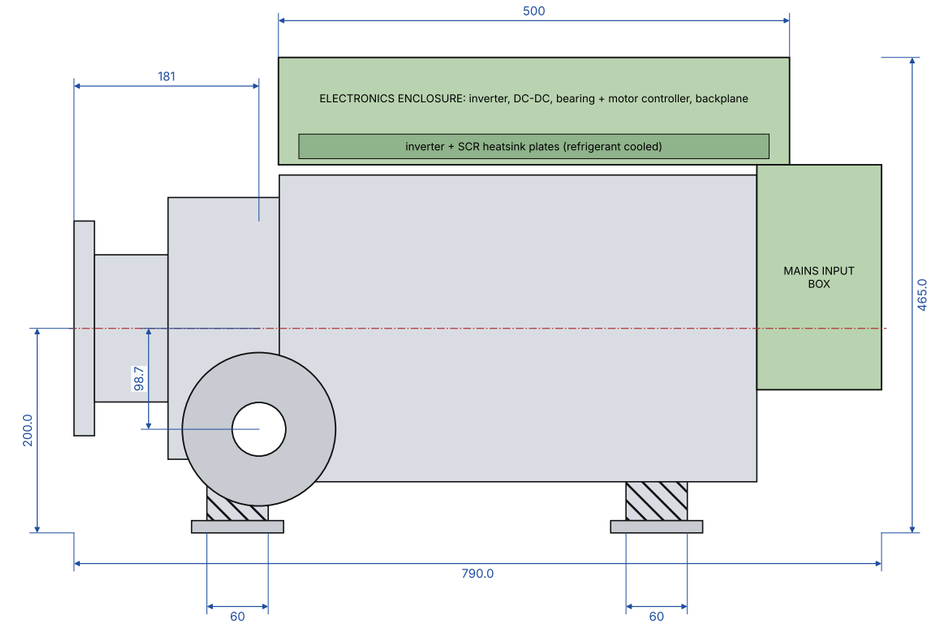
Motor cooling (IOM): liquid R-513A → solenoids/orifices → inverter & SCR heatsinks → stator grooves → vapour returns to suction through motor cavity.

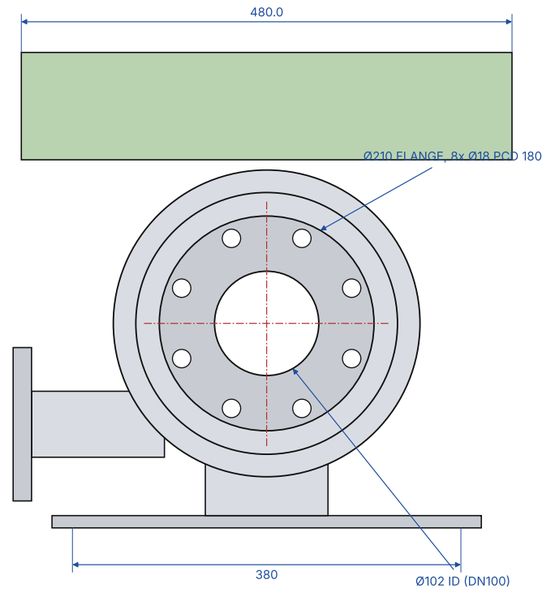
GENERAL ARRANGEMENT - SECTION

Turbocor-class 2-stage maglev centrifugal compressor (R-513A)	SHEET 2 OF 12
DWG TC-100/1 SCALE 1:3 UNITS mm	REV A 2026-10-17

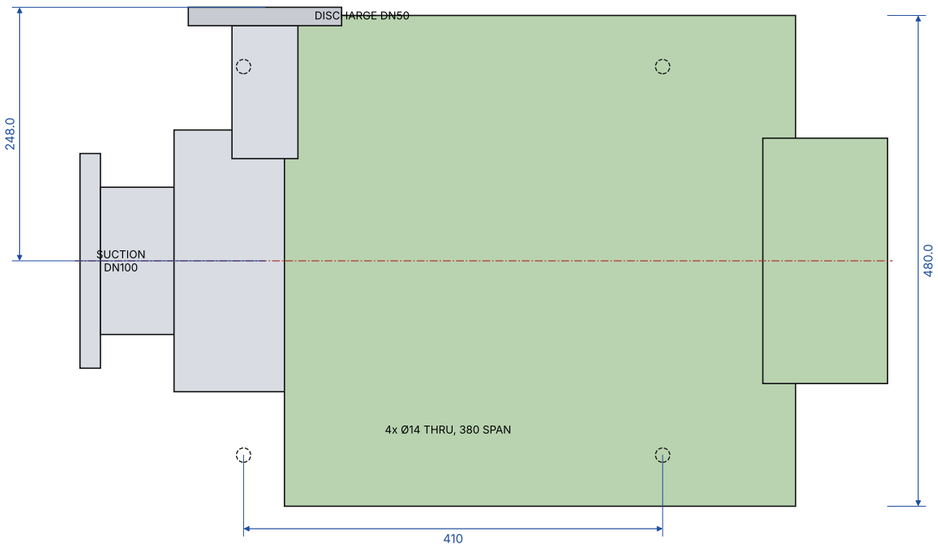
Reverse-engineered study model - NOT OEM geometry. Third-angle projection. General tolerance +/-0.5 mm unless noted.

SIDE ELEVATION 1:6





END VIEW FROM SUCTION 1:6 (+Y to the left)



PLAN 1:6

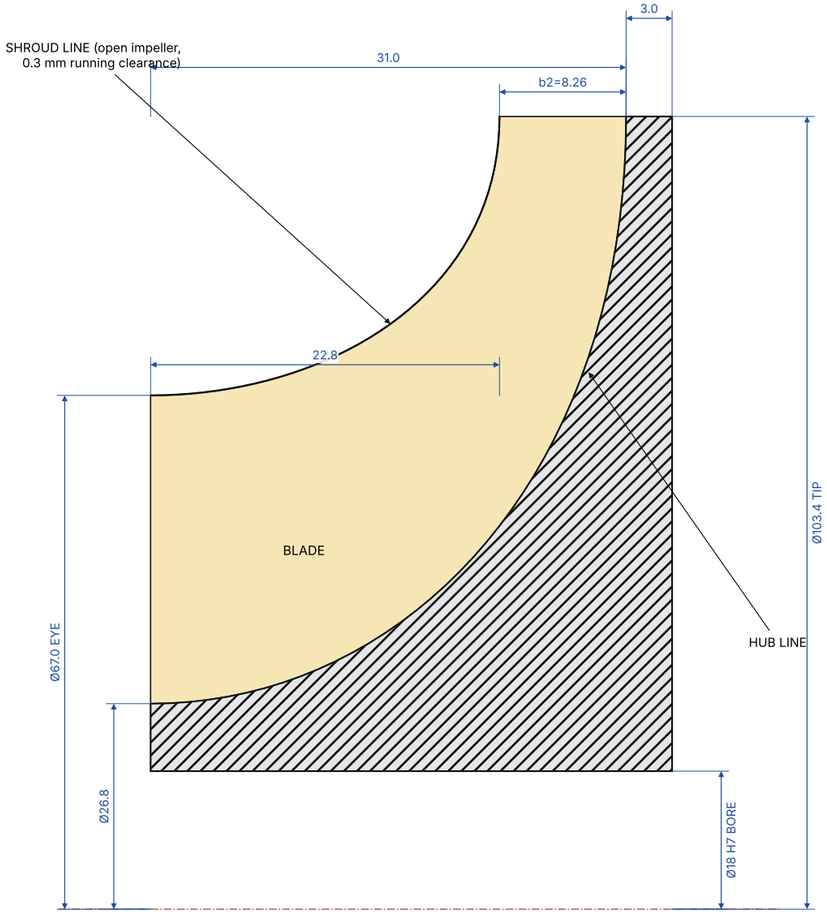
NOTES

1. Envelope ~790 × 480 × 465 mm matches published Turbocor TT-class dimensions.
2. Flange sizes from flow velocity: suction ~22 m/s, discharge ~21 m/s.
3. Electronics/mains boxes are simple sheet-metal shells - model as thin-wall parts.
4. CFD does not need the externals; suppress them in a 'CFD' configuration.

Reverse-engineered study model - NOT OEM geometry. Third-angle projection. General tolerance +/-0.5 mm unless noted.

GENERAL ARRANGEMENT - EXTERIOR

Turbocor-class 2-stage maglev centrifugal compressor (R-513A)	SHEET 3 OF 12
DWG TC-100/2 SCALE 1:6 UNITS mm	REV A 2026-10-17



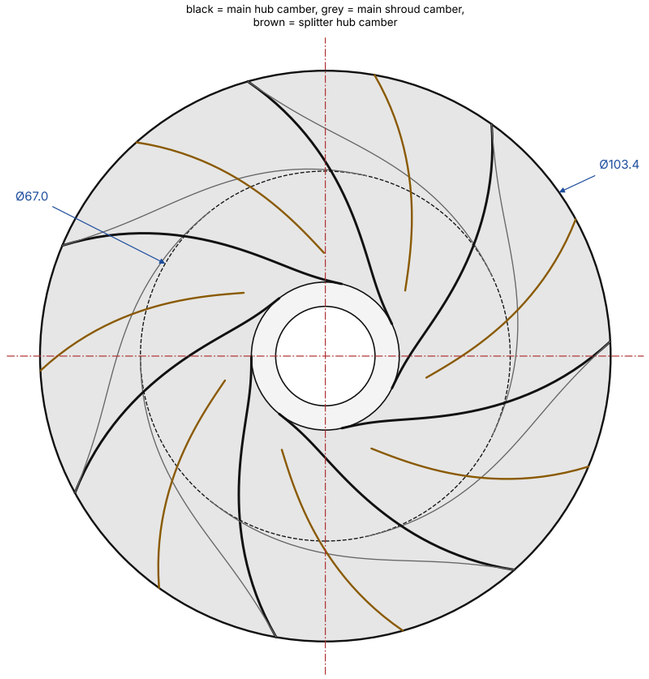
MERIDIONAL HALF-SECTION 2.5:1

BLADE DEFINITION (local z from eye, mm; β from meridional, θ wrap, deg)

m	hub z	hub r	hub β	hub θ	shr z	shr r	shr β	shr θ
0.00	0.0	13.4	34.2	-0.0	0.0	33.5	59.6	-0.0
0.11	6.2	14.1	31.1	-16.9	4.2	33.8	63.2	-13.4
0.23	12.0	16.2	28.8	-30.5	8.4	34.7	64.5	-27.9
0.34	17.3	19.5	27.5	-41.3	12.2	36.1	63.6	-41.8
0.46	21.9	23.9	27.4	-50.2	15.6	38.2	60.7	-53.5
0.58	25.7	29.5	28.7	-57.9	18.6	40.7	56.4	-62.9
0.71	28.6	36.0	31.4	-65.1	20.8	43.9	51.1	-70.0
0.85	30.4	43.4	35.4	-72.4	22.2	47.5	45.4	-75.6
1.00	31.0	51.7	40.0	-80.1	22.8	51.7	40.0	-80.1

KEY DATA

Feature	Value
Main blades / splitters	7 main + 7 splitter
Splitter LE at	30 % meridional length
Blade thickness hub / shroud	2.0 / 1.0 mm, LE elliptical 2:1
Exit blade angle β2b	40 deg backswept
Inlet relative Mach (tip)	0.88
Tip speed U2	162 m/s @ 30,000 rpm
Stage pressure ratio	1.979
Fillets	R1.0 hub, blade tips sharp
Curve files	cad_curves/stage1/*.txt



ROTATION: CW VIEWED FROM INLET (+X right-hand)

VIEW FROM INLET 0.9:1

NOTES

1. Machine from solid. Place eye plane at X = 95.0 in the assembly.
2. Blade = Boundary Surface between hub & shroud camber curves, Thicken 1.5 mm midplane.
3. Extend blade surfaces 1 mm past hub/shroud before Thicken, then trim (see guide).
4. Pattern blade 7x about axis; splitter pattern 7x offset by 25.71 deg.
5. Back face flat; balance holes optional. Tie-bolt clamps both impellers + spacer.

MATERIAL: 7075-T6 aluminium (or 2618-T61), balance G1.0, overspeed 115%

IMPELLER - STAGE 1

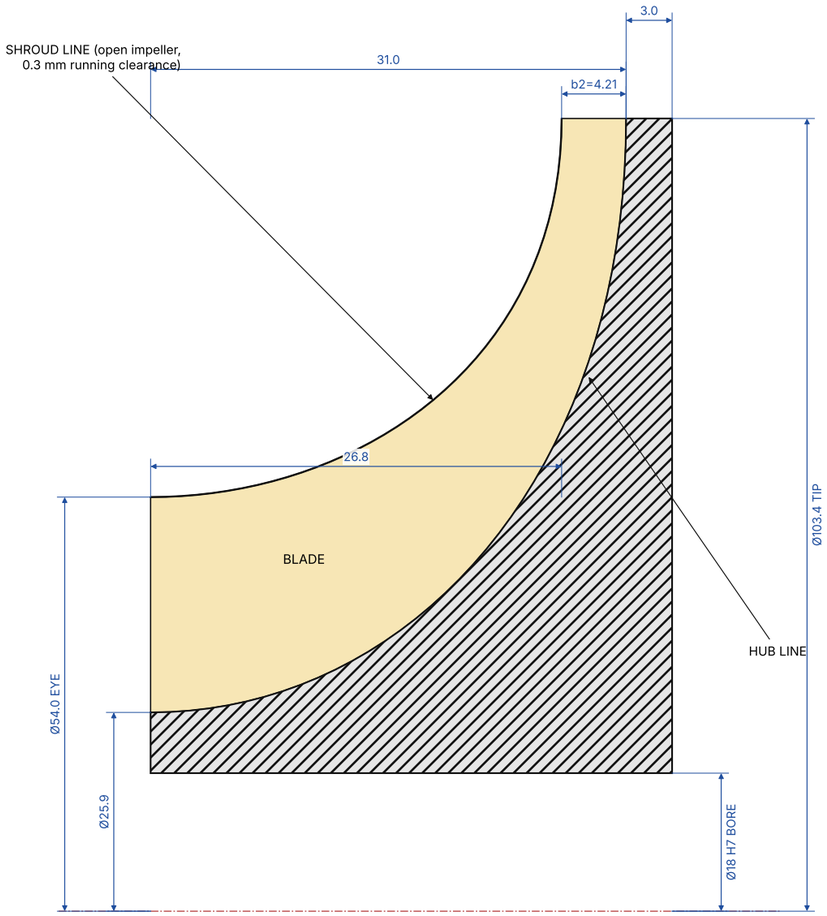
Turbocor-class 2-stage maglev centrifugal compressor (R-513A)

SHEET 4 OF 12

DWG TC-101 SCALE 2.5:1 / 0.9:1 UNITS mm

REV A 2026-10-17

Reverse-engineered study model - NOT OEM geometry. Third-angle projection. General tolerance +/-0.5 mm unless noted.



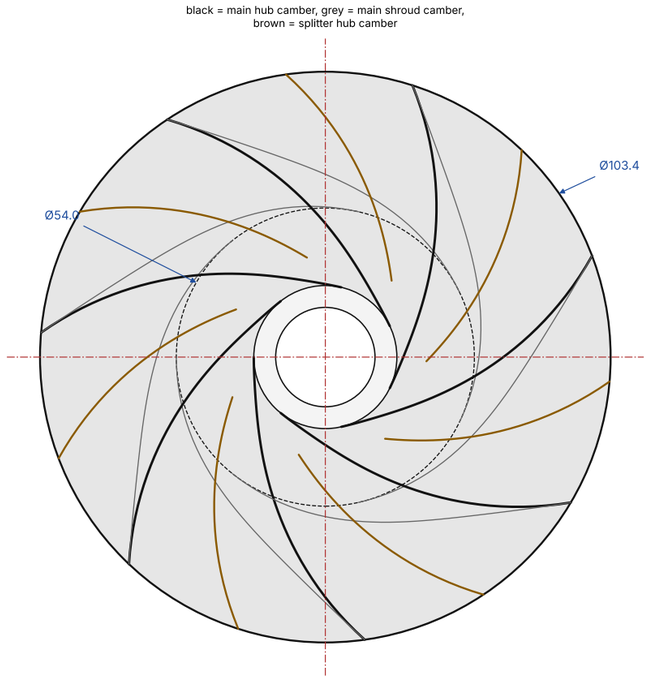
MERIDIONAL HALF-SECTION 2.5:1

BLADE DEFINITION (local z from eye, mm; β from meridional, θ wrap, deg)

m	hub z	hub r	hub β	hub θ	shr z	shr r	shr β	shr θ
0.00	0.0	13.0	37.8	-0.0	0.0	27.0	58.4	-0.0
0.11	6.2	13.7	36.0	-20.5	5.0	27.4	59.9	-17.8
0.23	12.0	15.7	34.5	-37.6	9.8	28.6	59.8	-35.5
0.34	17.3	19.1	33.5	-51.6	14.4	30.6	58.2	-51.6
0.46	21.9	23.6	33.3	-63.1	18.4	33.3	55.5	-65.1
0.58	25.7	29.2	33.8	-72.9	21.9	36.8	51.9	-76.1
0.71	28.6	35.8	35.1	-81.6	24.5	41.1	47.7	-84.9
0.85	30.4	43.4	37.3	-89.8	26.2	46.0	43.5	-92.0
1.00	31.0	51.7	40.0	-97.9	26.8	51.7	40.0	-97.9

KEY DATA

Feature	Value
Main blades / splitters	7 main + 7 splitter
Splitter LE at	30 % meridional length
Blade thickness hub / shroud	2.0 / 1.0 mm, LE elliptical 2:1
Exit blade angle β2b	40 deg backswept
Inlet relative Mach (tip)	0.73
Tip speed U2	162 m/s @ 30,000 rpm
Stage pressure ratio	1.970
Fillets	R1.0 hub, blade tips sharp
Curve files	cad_curves/stage2/*.txt



ROTATION: CW VIEWED FROM INLET (+X right-hand)

VIEW FROM INLET 0.9:1

NOTES

1. Machine from solid. Place eye plane at X = 152.0 in the assembly.
2. Blade = Boundary Surface between hub & shroud camber curves, Thicken 1.5 mm midplane.
3. Extend blade surfaces 1 mm past hub/shroud before Thicken, then trim (see guide).
4. Pattern blade 7x about axis; splitter pattern 7x offset by 25.71 deg.
5. Back face flat; balance holes optional. Tie-bolt clamps both impellers + spacer.

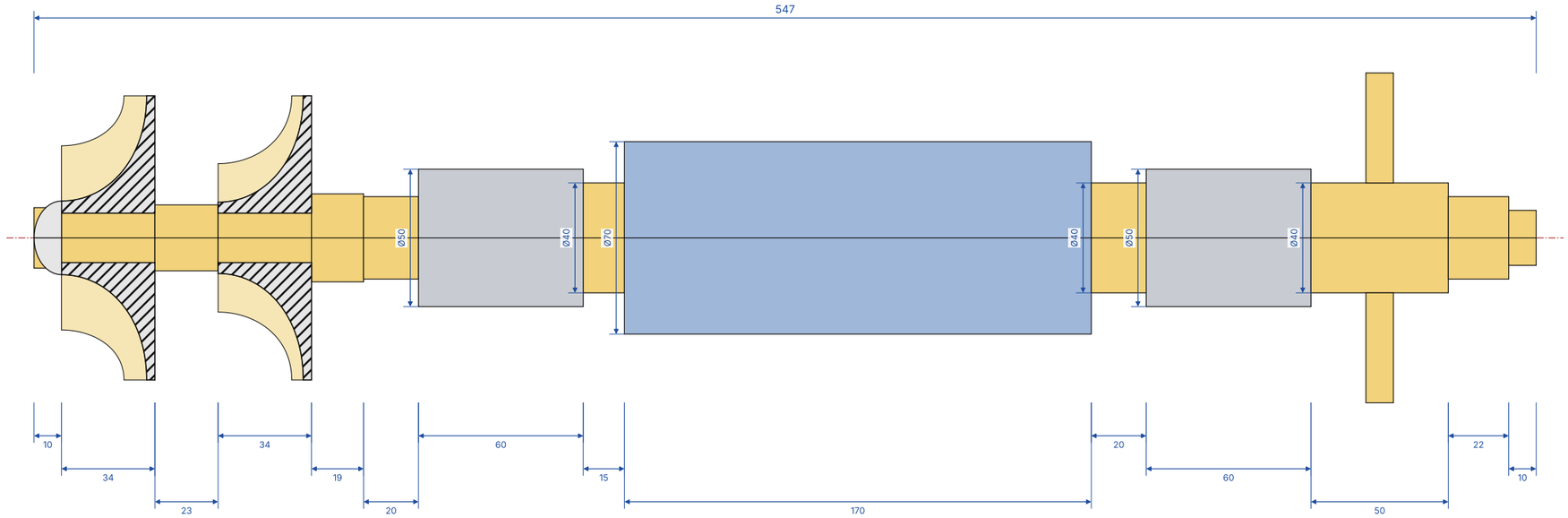
MATERIAL: 7075-T6 aluminium (or 2618-T61), balance G1.0, overspeed 115%

IMPELLER - STAGE 2

Turbocor-class 2-stage maglev centrifugal compressor (R-513A)	SHEET 5 OF 12
DWG TC-102 SCALE 2.5:1 / 0.9:1 UNITS mm	REV A 2026-10-17

Reverse-engineered study model - NOT OEM geometry. Third-angle projection. General tolerance +/-0.5 mm unless noted.

ROTOR ELEVATION 1:2



SEGMENT TABLE (X in machine coordinates)

Seg	X from	X to	Ø	Function
1	85.0	95.0	22	impeller nut / nose cone
2	95.0	129.0	18	stage-1 impeller seat
3	129.0	152.0	24	interstage spacer sleeve
4	152.0	186.0	18	stage-2 impeller seat
5	186.0	205.0	32	seal sleeve
6	205.0	225.0	30	front touchdown bearing journal
7	225.0	285.0	50	front radial bearing rotor laminations
8	285.0	300.0	40	shaft
9	300.0	470.0	70	PM motor rotor (magnets + carbon-fibre sleeve)
10	470.0	490.0	40	shaft
11	490.0	550.0	50	rear radial bearing rotor laminations
12	550.0	600.0	40	thrust-disk seat
13	600.0	622.0	30	rear touchdown bearing journal
14	622.0	632.0	20	axial / speed sensor target
TD	570	580	120	thrust disk (shrink fit, 17-4PH / Co-Fe)

NOTES

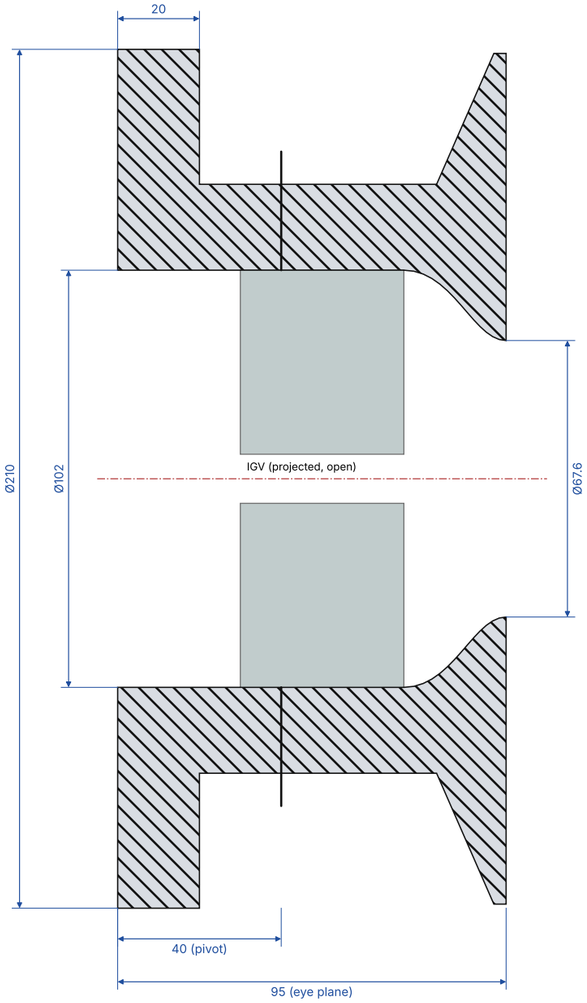
1. Estimated rotor mass ~9.0 kg (steel est. + 2 Al impellers).
2. Model as ONE revolved part from the segment table, then split bodies for colour.
3. Impellers, spacer sleeve & nose nut clamped by an M10 tie-bolt (not shown).
4. Motor: 2-pole surface PM, Sm2Co17 magnets, 5 mm CFRP sleeve (tip speed ~110 m/s).
5. Radial air gap at mag-bearing journals 0.5 mm; touchdown clearance 0.2 mm radial.
6. First bending critical must stay >20% above 39,000 rpm max - check in Ansys Modal.

MATERIAL: 17-4PH H1025 shaft; CFRP magnet sleeve; Si-steel/Co-Fe bearing laminations

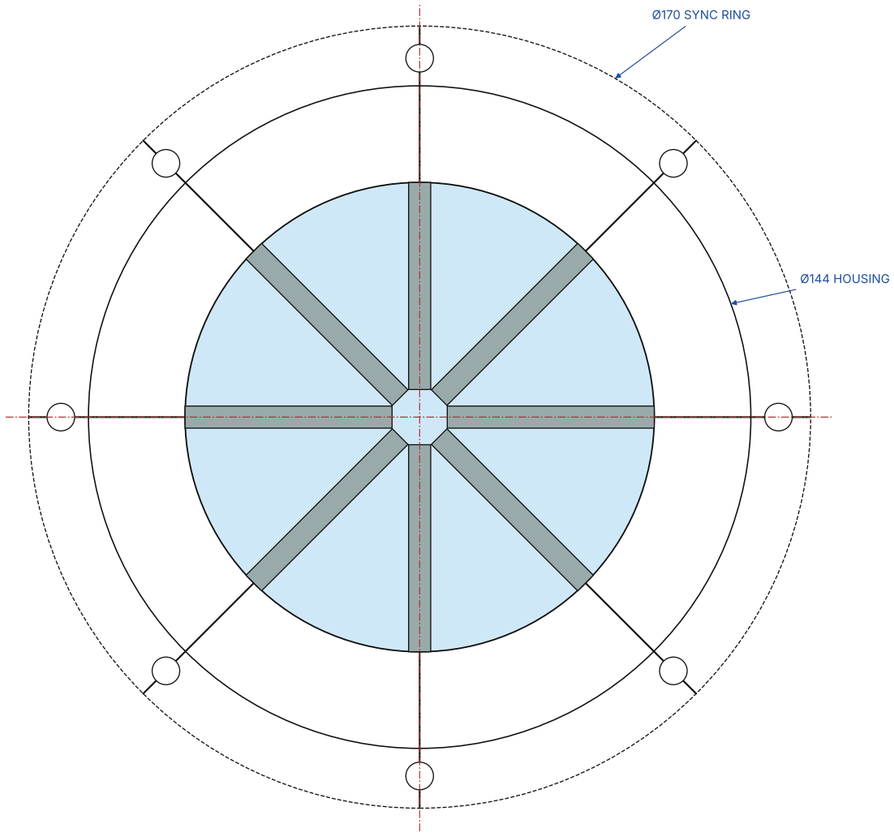
ROTOR SHAFT ASSEMBLY

Turbocor-class 2-stage maglev centrifugal compressor (R-513A)	SHEET 6 OF 12
DWG TC-103 SCALE 1:2 UNITS mm	REV A 2026-10-17

Reverse-engineered study model - NOT OEM geometry. Third-angle projection. General tolerance +/-0.5 mm unless noted.

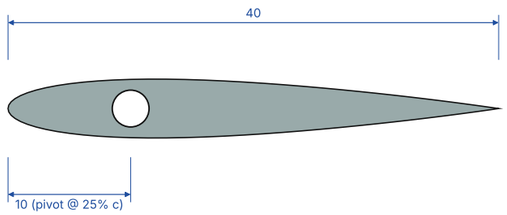


INLET SECTION 1:1.5



IGV ASSEMBLY - VIEW FROM SUCTION 3:4 (vanes at 0° = fully open)

VANE SECTION NACA 0012, chord 40 2:1



NOTES

1. 8 cantilevered vanes, span r = 6 to 51 mm, pivot pin Ø6 through housing wall.
2. Range 0 deg (open) to 70 deg closed; prewhirl in rotation direction reduces capacity.
3. For CFD at design point model IGVs at 0 deg or omit them (they are a pressure-drop item).
4. Profile ready to revolve: cad_curves/profiles/inlet_housing.dxf (Front plane, X axis).

MATERIAL: Housing A356-T6; vanes 6061-T6; pivot pins 316 SS w/ PTFE bushes

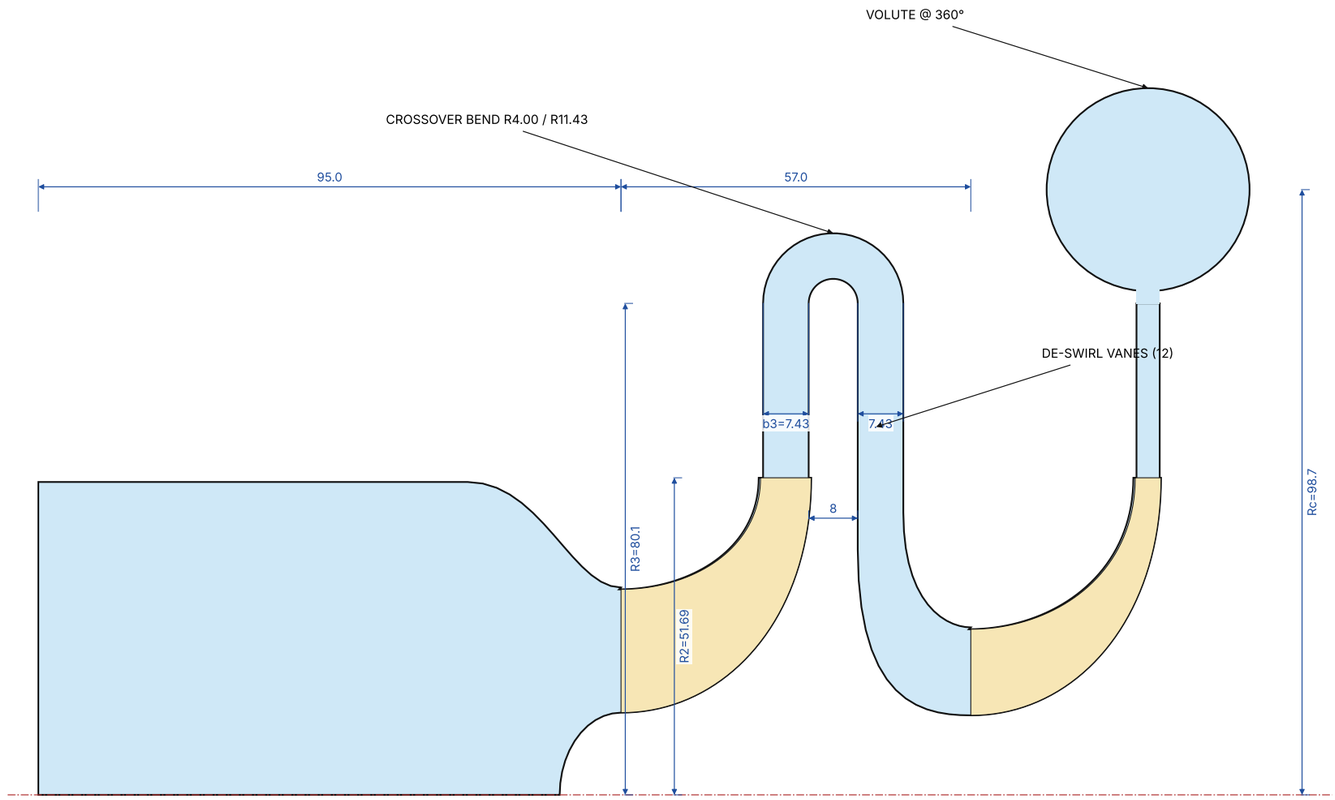
INLET HOUSING & INLET GUIDE VANES

Turbocor-class 2-stage maglev centrifugal compressor (R-513A)	SHEET 7 OF 12
DWG TC-110 SCALE 1:1.5 / 3:4 / 2:1 UNITS mm	REV A 2026-10-17

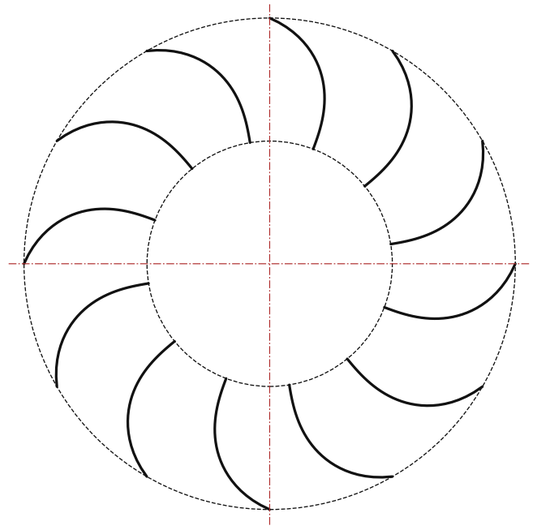
Reverse-engineered study model - NOT OEM geometry. Third-angle projection. General tolerance +/-0.5 mm unless noted.

FLOW-PATH STATIONS (mm)

Plane	X	r hub	r shroud
Suction flange	0.0	0.00	51.00
IGV pivot	40.0	0.00	51.00
Eye 1	95.0	13.40	33.50
Diffuser 1 in	118.17-125.60	51.69	51.69
Diffuser 1 out	118.17-125.60	80.11	80.11
Return ch. in	133.60-141.03	80.11	80.11
Eye 2	152.0	12.96	27.00
Diffuser 2 out	179.05-182.83	80.11	80.11



MERIDIONAL FLOW PATH (upper half) 1:1



DE-SWIRL VANES - plan 1:2

De-swirl vane	Value
Count	12
Inlet r / metal angle	80.1 / 66.6° from radial
Exit r / metal angle	40 / 0° (no swirl)
Thickness	3 mm, round LE, sharp TE
Camber law	β = β_in (1-f)^1.5

NOTES

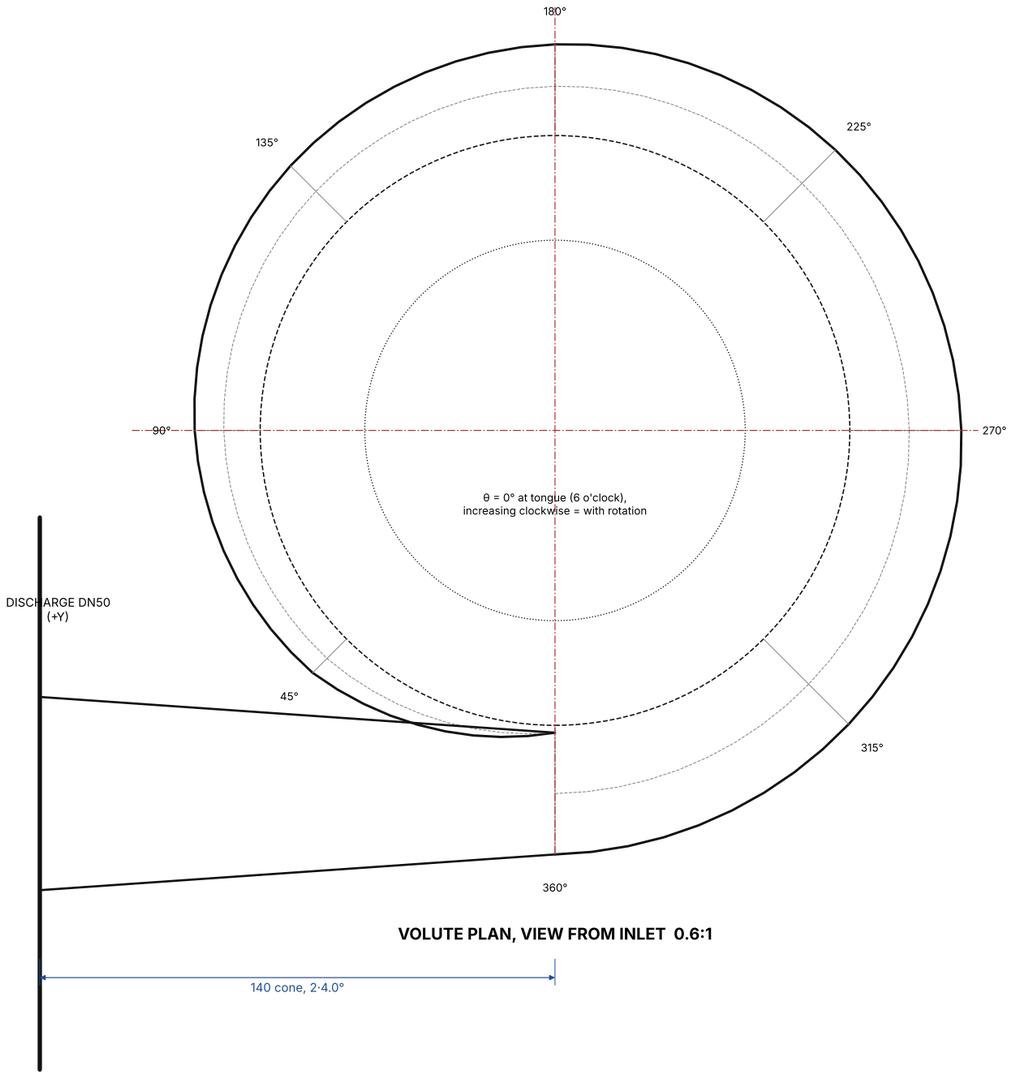
1. Closed profiles: cad_curves/profiles/*.dxf (casing, diaphragm, fluid) - import onto
2. the Front plane and Revolve about the X axis. Pieces: cad_curves/flowpath/.
3. Split the casting at diffuser-1 hub wall (diaphragm) & diffuser-2 hub wall (bearing housing).
4. Diaphragm is held by the de-swirl vanes; it carries the interstage labyrinth at Ø26.

MATERIAL: A356-T6 casting (split: inlet/shroud casing, diaphragm, volute casing)

COMPRESSOR CASING FLOW PATH & DE-SWIRL VANES

Turbocor-class 2-stage maglev centrifugal compressor (R-513A)	SHEET 8 OF 12
DWG TC-120 SCALE 1:1 / 1:2 UNITS mm	REV A 2026-10-17

Reverse-engineered study model - NOT OEM geometry. Third-angle projection. General tolerance +/-0.5 mm unless noted.



VOLUTE PLAN, VIEW FROM INLET 0.6:1

VOLUTE SECTIONS (circular, external)

θ deg	Ø section	Rc	Area mm²
0	0.0	82.1	0
45	11.0	87.6	95
90	15.8	90.0	196
135	19.6	91.9	300
180	22.8	93.5	407
225	25.7	94.9	517
270	28.3	96.3	629
315	30.8	97.5	743
360	33.1	98.7	860

NOTES

1. Sizing: constant angular momentum, A(θ)/Rc = Q·θ / (2π·Cθ2·R2).
2. Q at volute = 45.9 L/s, Cθ2 = 102 m/s at R2 = 51.7 mm.
3. Model: Sweep a circle along the Rc spiral with a guide curve on the outer edge, OR
4. Loft through the 9 circular sections (planes every 45°) - easier in SolidWorks.
5. Tongue clearance 5% of R3. Section plane = through rotor axis.
6. Volute axial centre X = 180.94 (mid-width of diffuser 2).

Reverse-engineered study model - NOT OEM geometry. Third-angle projection. General tolerance +/-0.5 mm unless noted.

MATERIAL: A356-T6, integral with stage-2 casing

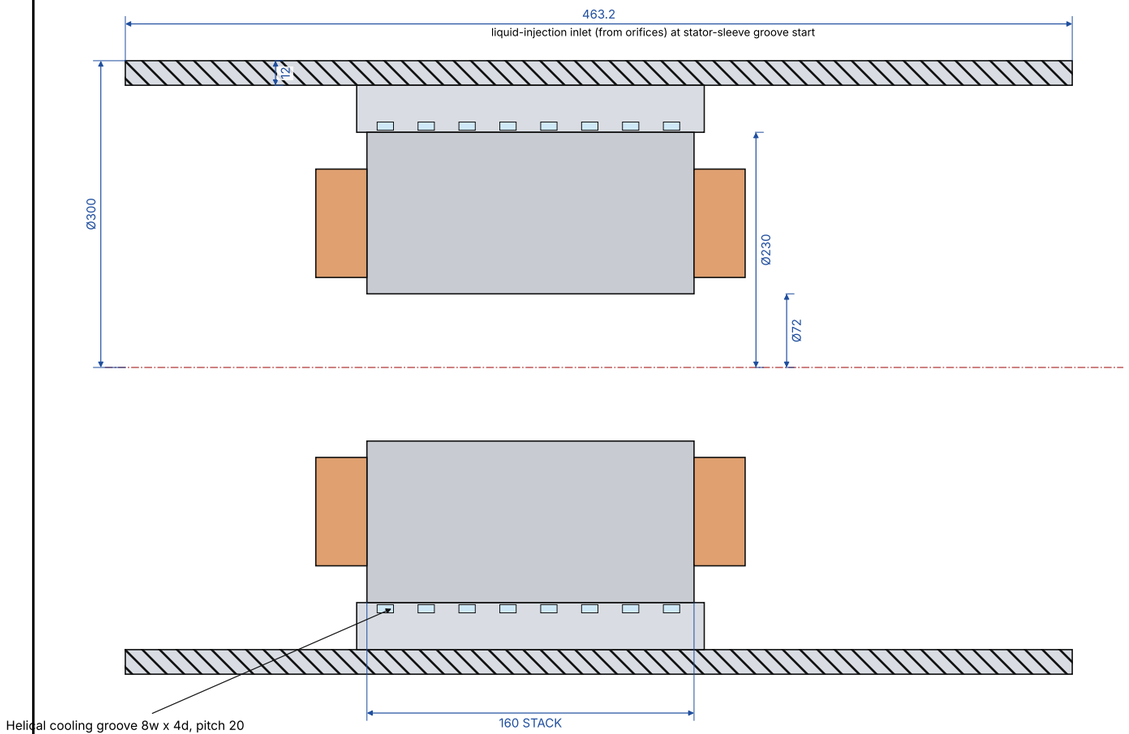
VOLUTE

Turbocor-class 2-stage maglev centrifugal compressor (R-513A)

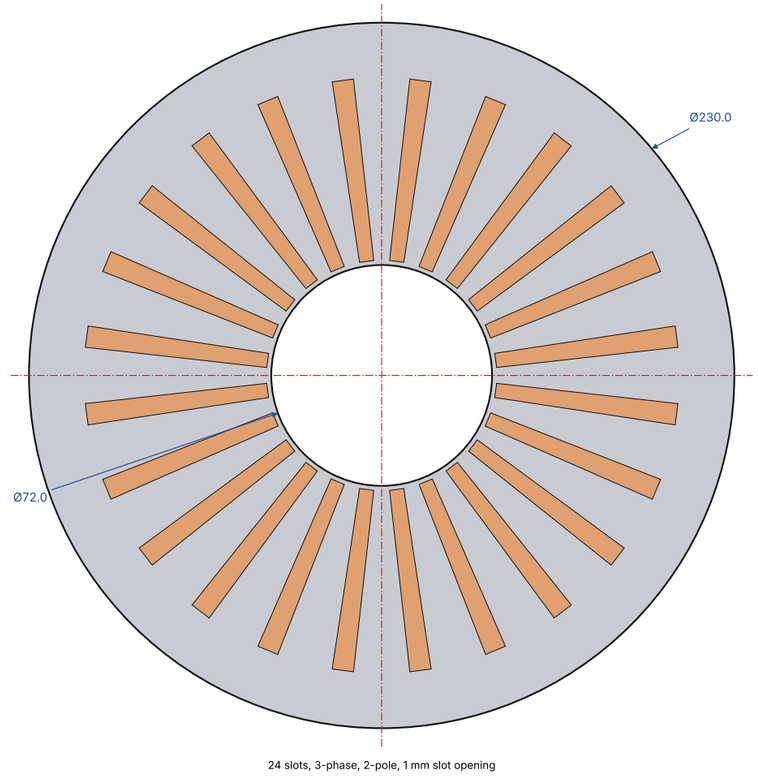
SHEET 9 OF 12

DWG TC-121 SCALE 0.6:1 UNITS mm

REV A 2026-10-17



HOUSING SECTION 1:3



STATOR LAMINATION 1:2

NOTES

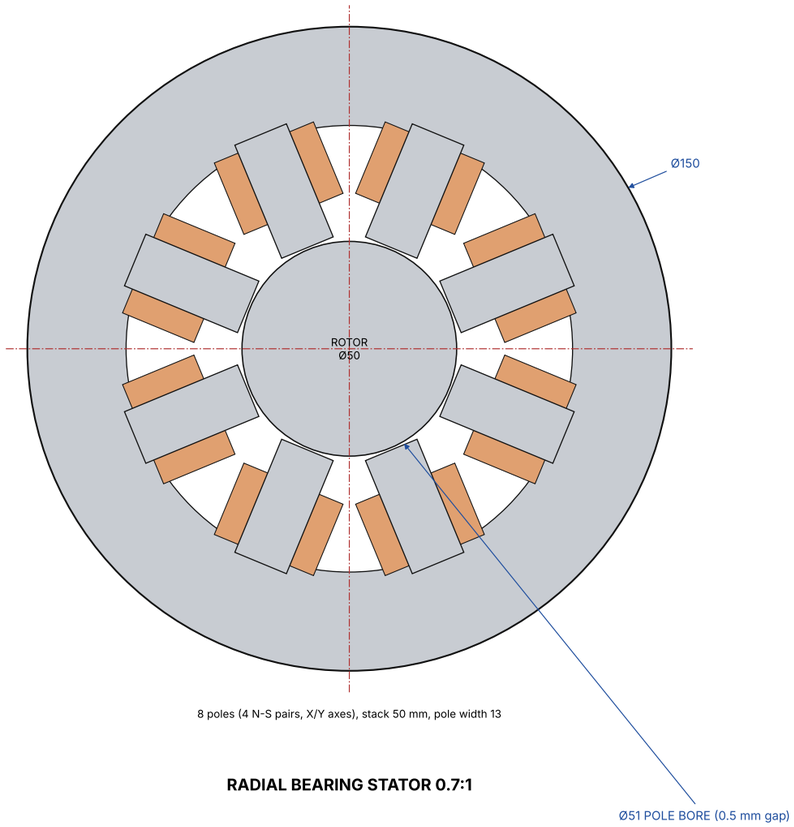
1. Rated ~113 kW at 30,000 rpm (36 N.m); 500 Hz electrical at 30k rpm (2-pole).
2. Air gap 1 mm to CFRP sleeve (rotor Ø70 over sleeve, stator bore Ø72).
3. Housing: revolve the section, cut helical groove with Swept Cut (Helix/Spiral, pitch 20).
4. Stator stack is press-fit into the groove sleeve; end windings modelled as simple rings.
5. Motor cavity is vented to suction - for CFD this region is NOT in the main gas path.

Reverse-engineered study model - NOT OEM geometry. Third-angle projection. General tolerance +/-0.5 mm unless noted.

MATERIAL: Housing A356-T6 / ductile iron; stator 0.2 mm NO20 laminations; Cu Litz windings

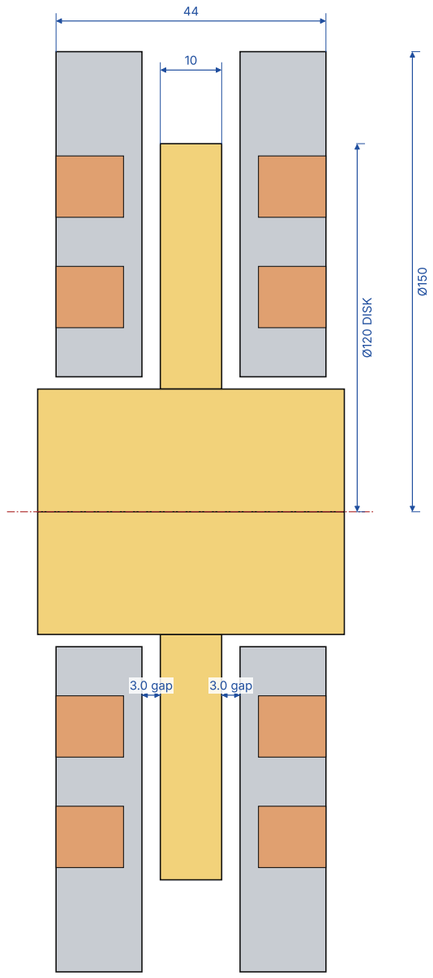
MOTOR STATOR & HOUSING

Turbocor-class 2-stage maglev centrifugal compressor (R-513A)	SHEET 10 OF 12
DWG TC-130 SCALE 1:3 / 1:2 UNITS mm	REV A 2026-10-17

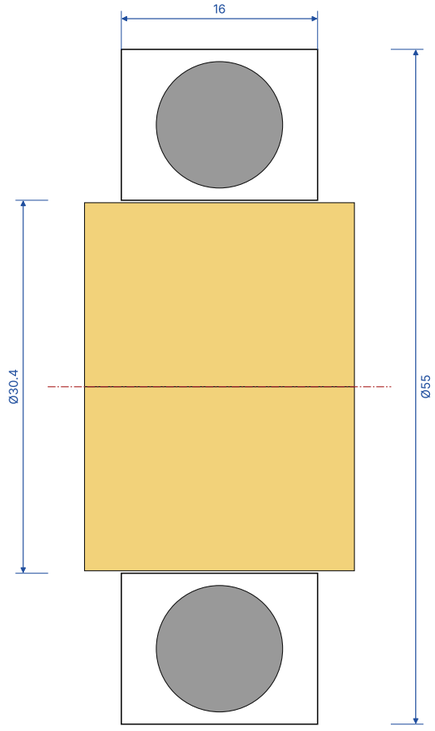


8 poles (4 N-S pairs, X/Y axes), stack 50 mm, pole width 13

RADIAL BEARING STATOR 0.7:1



AXIAL BEARING SECTION 1:1



TOUCHDOWN BEARING 2:1

NOTES

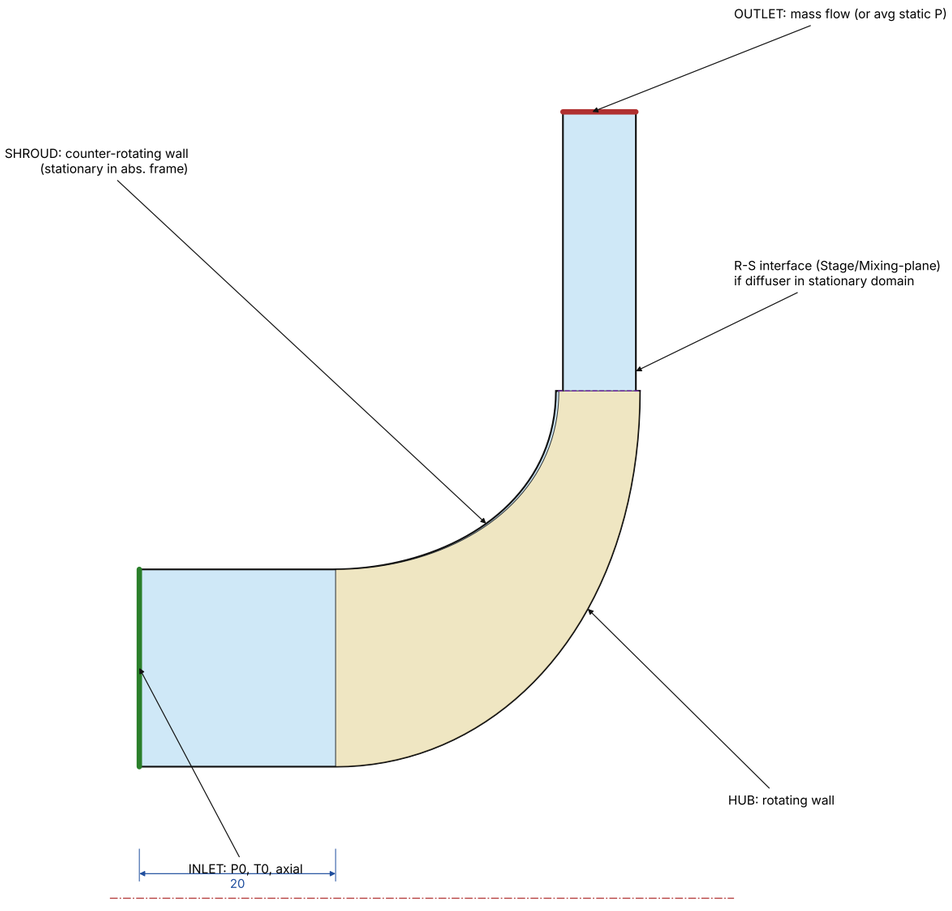
1. Bearings are electromagnets - for CAD model laminations/coils as simple solids.
2. Radial: 0.5 mm air gap; touchdown (ceramic hybrid angular-contact pair): 0.2 mm radial
3. clearance, so the rotor lands on touchdowns before touching magnetic poles.
4. Axial: two stators straddle the thrust disk, 0.5 mm nominal gap each side.
5. Position sensors (eddy-current) sit next to each radial bearing and at the shaft end.

Reverse-engineered study model - NOT OEM geometry. Third-angle projection. General tolerance +/-0.5 mm unless noted.

MATERIAL: Stators: laminated Si-steel (radial), solid Co-Fe / low-C steel (axial); Cu coils

MAGNETIC & TOUCHDOWN BEARINGS

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DOMAIN A: STAGE-1 SINGLE PASSAGE (TurboGrid → CFX) 1.6:1

DOMAIN A SETUP

Item	Setting
Mass flow (full wheel)	3.343 kg/s
Per passage (7 main+splitter)	0.4776 kg/s
Inlet P0 / T0	364.3 kPa / 279.65 K
Rotation	30,000 rpm about +X
Periodicity	360/7 = 51.43 deg rotational
Tip clearance	0.3 mm at shroud
Turbulence	SST k-ω, y+ ~1 on blades
Fluid	R-513A real gas (RGP / Peng-Robinson)
Expected stage-1 PR	~1.98 (meanline)

NOTES

1. DOMAIN B (full-stage / 360 deg, Fluent or CFX): extract the fluid volume of the assembly
2. (SpaceClaim Volume Extract) from suction flange to discharge flange, rotor stationary
3. domains split at impeller inlet/outlet; Frozen Rotor interfaces; ~10-20 M cells.
4. Run DOMAIN A first (minutes-hours on a workstation); B only once A is converged and sane.
5. Do NOT mesh the motor, bearings, or electronics for the aero CFD.
6. Speed lines: vary back-pressure/mass flow at 24k, 27k, 30k, 33k rpm to map surge/choke.
7. Post: PR_tt, η_tt (from h0 or T0/P0 with real gas), blade loading, Mach number at 50% span.

Reverse-engineered study model - NOT OEM geometry. Third-angle projection. General tolerance +/-0.5 mm unless noted.

CFD FLUID DOMAINS & BOUNDARY CONDITIONS

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